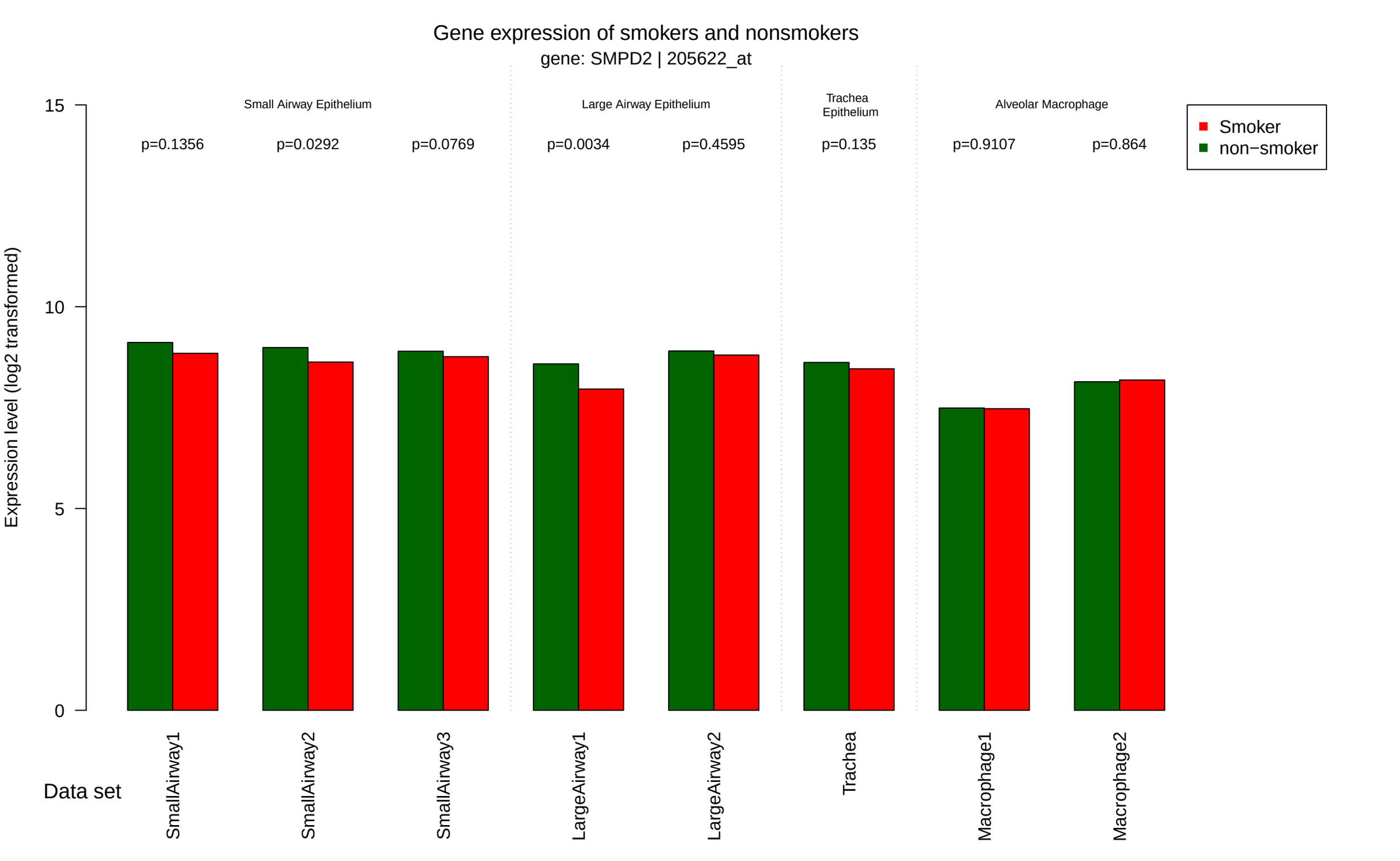
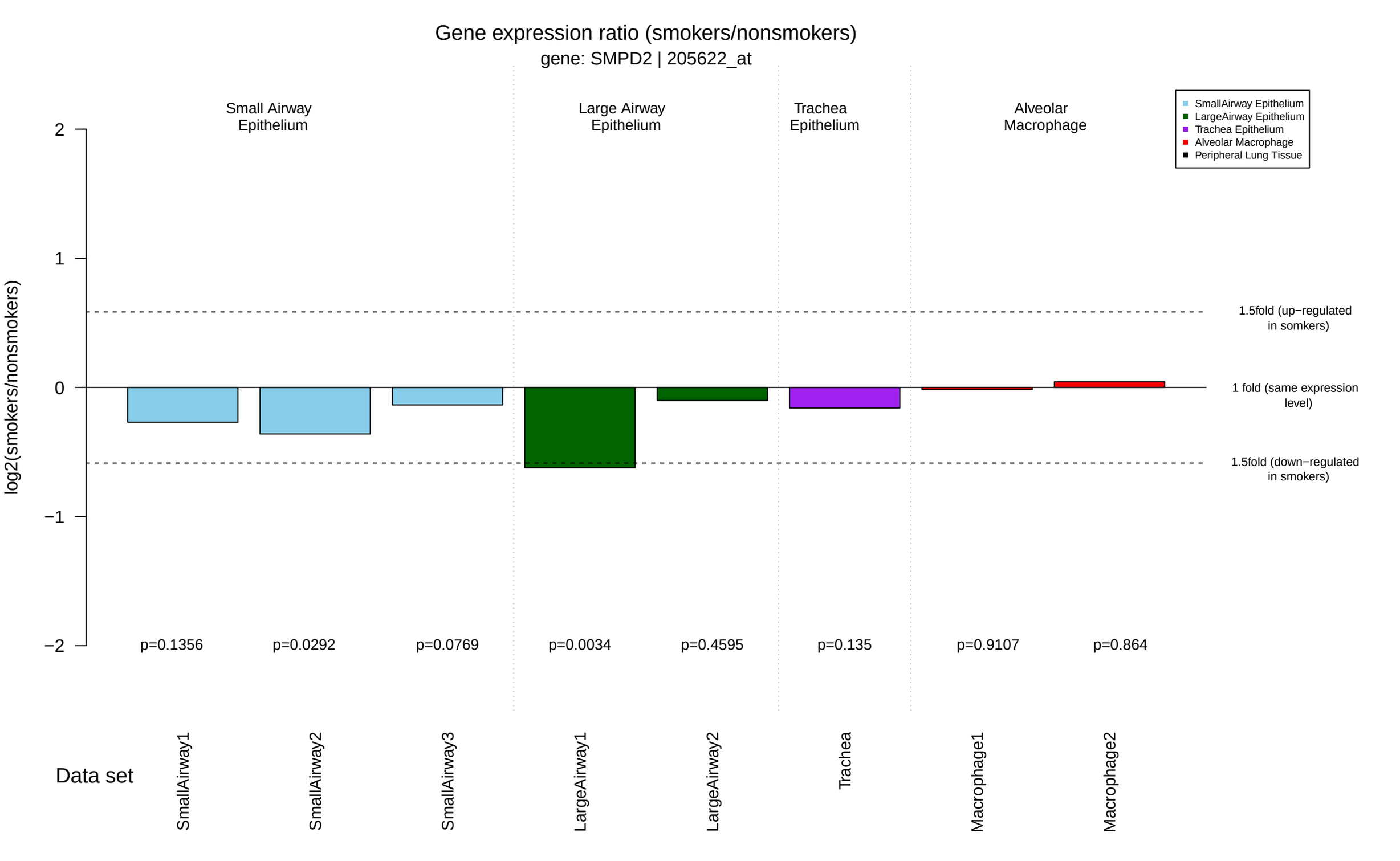
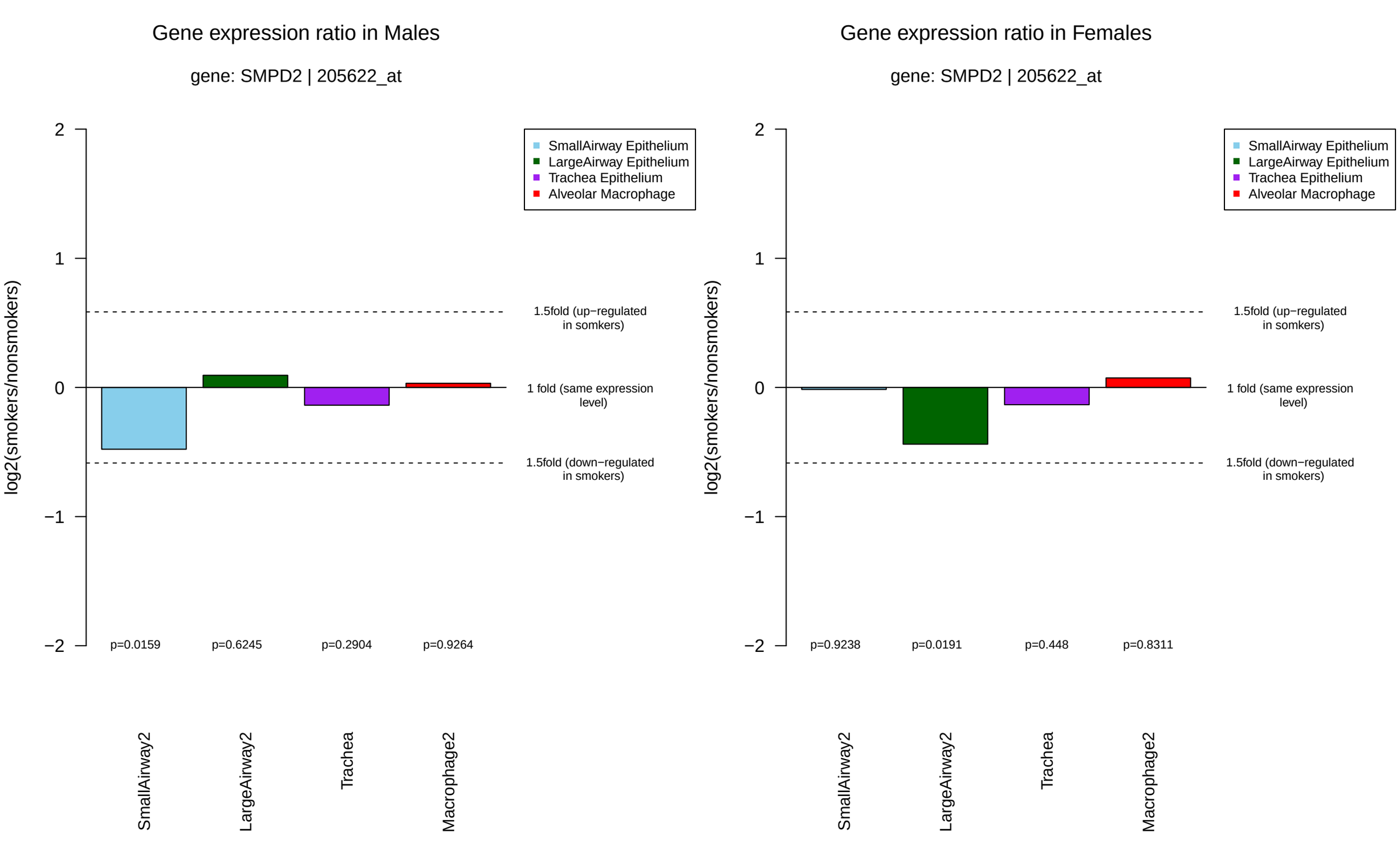
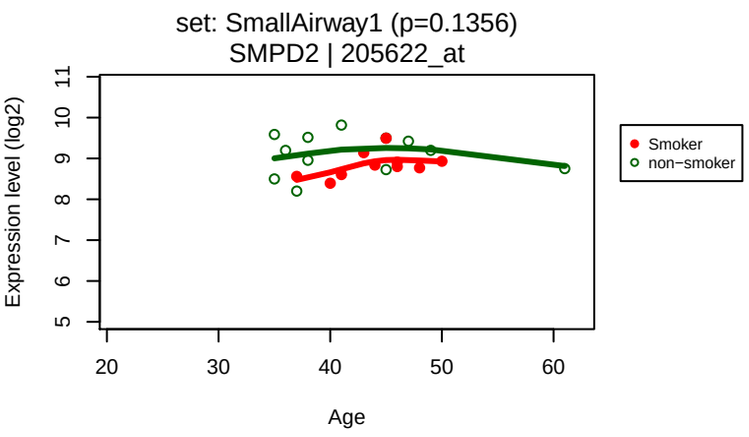
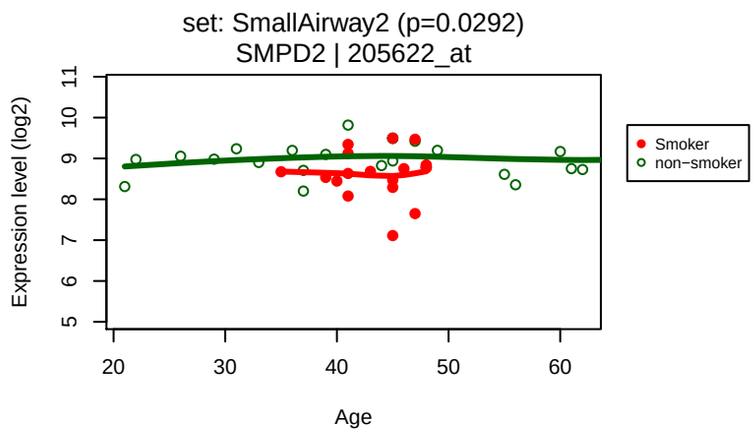
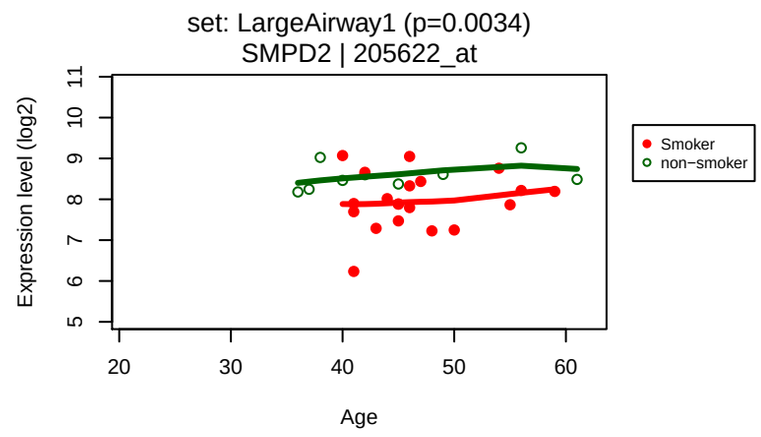

Gene expression of smokers and nonsmokers
gene: SMPD2 | 205622_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1356
p=0.0292
p=0.0769
p=0.0034
p=0.4595
p=0.135
p=0.9107
p=0.864
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

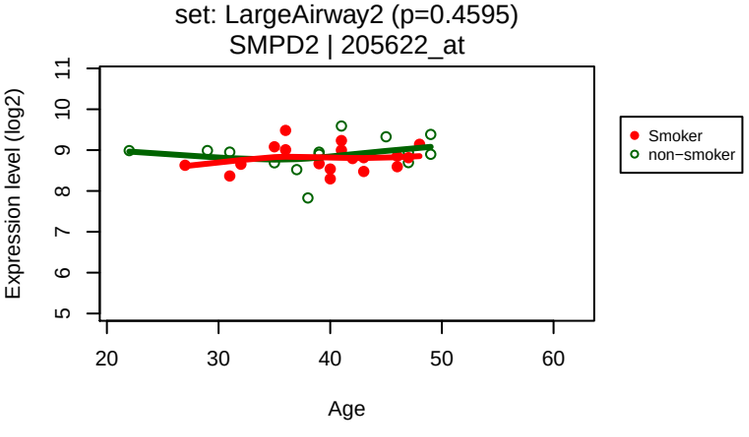

Gene expression ratio (smokers/nonsmokers)
gene: SMPD2 | 205622_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1356
p=0.0292
p=0.0769
p=0.0034
p=0.4595
p=0.135
p=0.9107
p=0.864
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

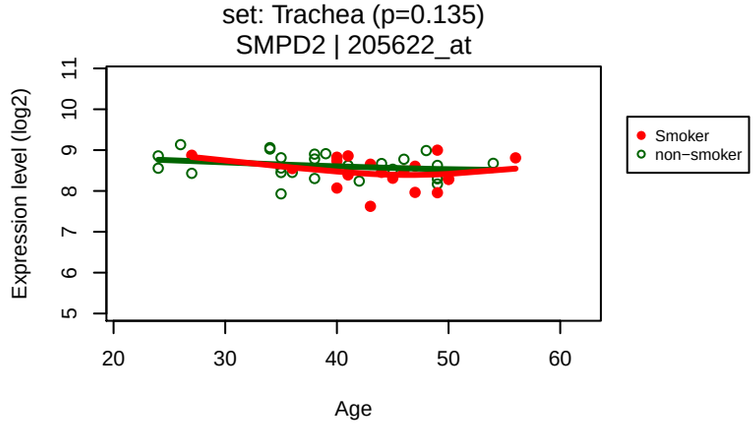

Gene expression ratio in Males
gene: SMPD2 | 205622_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0159
p=0.6245
p=0.2904
p=0.9264
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMPD2 | 205622_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9238
p=0.0191
p=0.448
p=0.8311
SmallAirway2
LargeAirway2
Trachea
Macrophage2

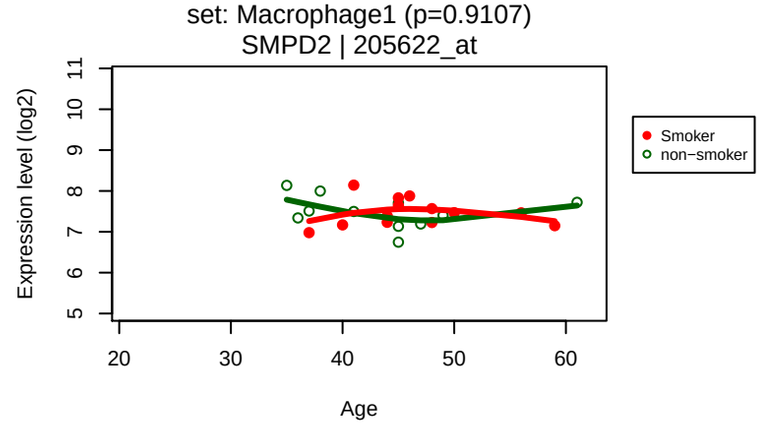

set: SmallAirway1 (p=0.1356)
SMPD2 | 205622_at
Expression level (log2)
Age
Smoker
non−smoker

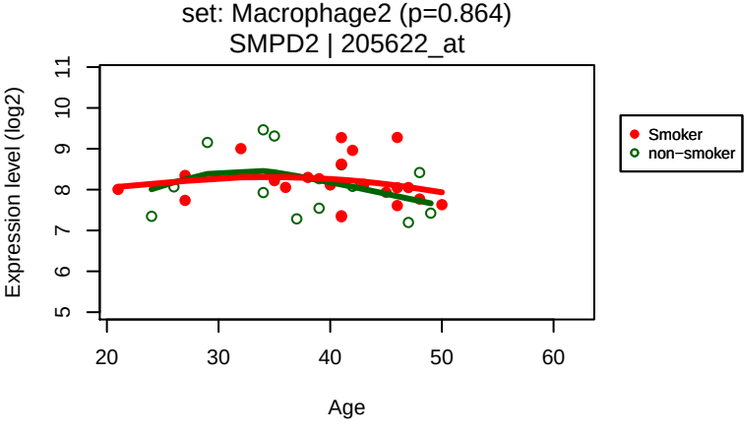

set: SmallAirway2 (p=0.0292)
SMPD2 | 205622_at
Expression level (log2)
Age
Smoker
non−smoker

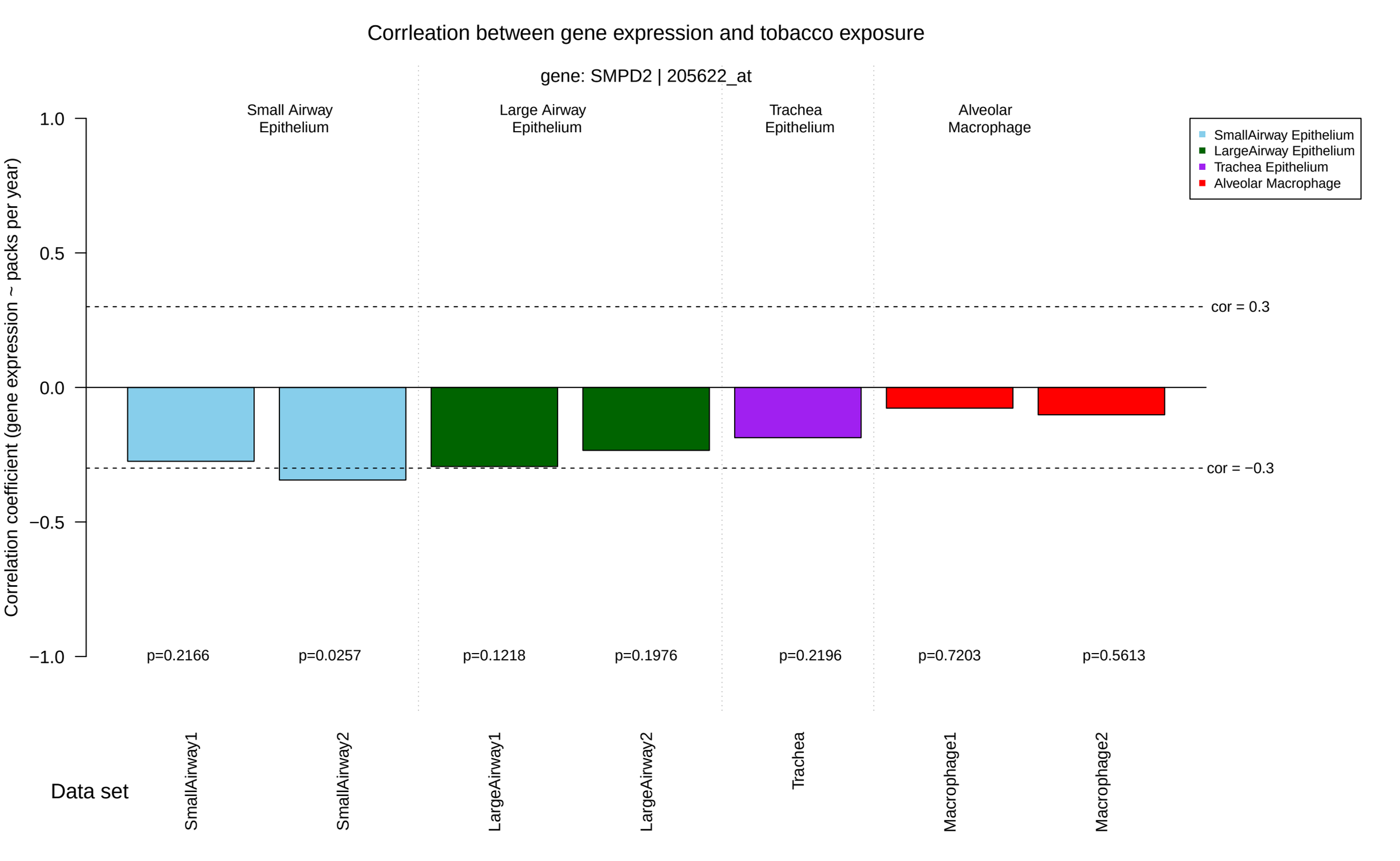

set: LargeAirway1 (p=0.0034)
SMPD2 | 205622_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4595)
SMPD2 | 205622_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.135)
SMPD2 | 205622_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9107)
SMPD2 | 205622_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.864)
SMPD2 | 205622_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMPD2 | 205622_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2166
p=0.0257
p=0.1218
p=0.1976
p=0.2196
p=0.7203
p=0.5613
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2